

# CS2601 Linear and Convex Optimization

## Homework 3 Solution

Due: 2021.10.18

1. Suppose  $f$  is a convex function and  $S \subset \text{dom } f$  is a convex set. Let  $M$  be the set of global minima of  $f$  over  $S$ ,

$$M = \{\mathbf{x}^* \in S : f(\mathbf{x}^*) \leq f(\mathbf{x}), \forall \mathbf{x} \in S\}.$$

Show that  $M$  is a convex set.

*Proof.* Let  $\mathbf{x}^*, \mathbf{y}^* \in M$  and  $\theta \in [0, 1]$ . Since  $S$  is convex,  $\theta\mathbf{x}^* + \bar{\theta}\mathbf{y}^* \in S$ . Since  $f$  is convex, for any  $\mathbf{x} \in S$ ,

$$f(\theta\mathbf{x}^* + \bar{\theta}\mathbf{y}^*) \leq \theta f(\mathbf{x}^*) + \bar{\theta} f(\mathbf{y}^*) \leq \theta f(\mathbf{x}) + \bar{\theta} f(\mathbf{x}) = f(\mathbf{x})$$

Thus  $\theta\mathbf{x}^* + \bar{\theta}\mathbf{y}^* \in M$  and  $M$  is convex. □

2. Let  $f$  be convex. If  $f(\theta\mathbf{x} + \bar{\theta}\mathbf{y}) = \theta f(\mathbf{x}) + \bar{\theta} f(\mathbf{y})$  for some  $\mathbf{x}, \mathbf{y}$  and  $\theta = \theta_0 \in (0, 1)$ , then it holds for the same  $\mathbf{x}, \mathbf{y}$  and any  $\theta \in [0, 1]$ .

Hint: Assume  $f(\theta_1\mathbf{x} + \bar{\theta}_1\mathbf{y}) < \theta_1 f(\mathbf{x}) + \bar{\theta}_1 f(\mathbf{y})$  for some  $\theta_1$ . Without loss of generality, you may assume  $\theta_1 \in (0, \theta_0)$ ; the case  $\theta_1 \in (\theta_0, 1)$  is similar. Express  $\theta_0\mathbf{x} + \bar{\theta}_0\mathbf{y}$  as a convex combination of  $\theta_1\mathbf{x} + \bar{\theta}_1\mathbf{y}$  and  $\mathbf{x}$ . Then deduce a contradiction.

*Proof.* Suppose  $f(\theta_1\mathbf{x} + \bar{\theta}_1\mathbf{y}) < \theta_1 f(\mathbf{x}) + \bar{\theta}_1 f(\mathbf{y})$  for some  $\theta_1 \in (0, \theta_0)$ . Let  $\mathbf{z}_i = \theta_i\mathbf{x} + \bar{\theta}_i\mathbf{y}$  for  $i = 0, 1$ . We want to express  $\mathbf{z}_0$  as a convex combination of  $\mathbf{x}$  and  $\mathbf{z}_1$ , i.e. we want to find  $\alpha$  s.t.

$$\alpha\mathbf{x} + \bar{\alpha}\mathbf{z}_1 = \alpha\mathbf{x} + \bar{\alpha}(\theta_1\mathbf{x} + \bar{\theta}_1\mathbf{y}) = \mathbf{z}_0 = \theta_0\mathbf{x} + \bar{\theta}_0\mathbf{y}$$

Comparing the coefficients of  $\mathbf{x}$  and  $\mathbf{y}$ , we see it suffices to set  $\bar{\alpha} = \bar{\theta}_0/\bar{\theta}_1 \in (0, 1)$ . Thus

$$\begin{aligned} f(\mathbf{z}_0) &= f(\alpha\mathbf{x} + \bar{\alpha}\mathbf{z}_1) \\ &\leq \alpha f(\mathbf{x}) + \bar{\alpha} f(\mathbf{z}_1) \quad \text{by convexity of } f \\ &< \alpha f(\mathbf{x}) + \bar{\alpha}[\theta_1 f(\mathbf{x}) + \bar{\theta}_1 f(\mathbf{y})] \quad \text{by assumption about } \mathbf{z}_1 \\ &= \theta_0 f(\mathbf{x}) + \bar{\theta}_0 f(\mathbf{y}) \quad \text{by definition of } \alpha \\ &= f(\mathbf{z}_0) \quad \text{by assumption about } \mathbf{z}_0 \end{aligned}$$

which is a contradiction, so  $f(\theta\mathbf{x} + \bar{\theta}\mathbf{y}) = \theta f(\mathbf{x}) + \bar{\theta} f(\mathbf{y})$  for all  $\theta \in (0, \theta_0)$ . A similar proof shows  $f(\theta\mathbf{x} + \bar{\theta}\mathbf{y}) = \theta f(\mathbf{x}) + \bar{\theta} f(\mathbf{y})$  for  $\theta \in (\theta_0, 1)$ . □

3. Determine if the following functions are convex, concave, or neither.

(a).  $f(\mathbf{x}) = f(x_1, x_2, x_3) = x_1^2 + x_1x_3 + x_2^2 + x_2x_3 + \frac{1}{2}x_3^2$  on  $\mathbb{R}^3$

(b).  $f(\mathbf{x}) = f(x_1, x_2) = (x_1x_2)^{-1}$  on  $\mathbb{R}_{++}^2 = \{(x_1, x_2) : x_1 > 0, x_2 > 0\}$

(c).  $f(x_1, x_2) = x_1x_2^2$  on  $\mathbb{R}_{++}^2 = \{(x_1, x_2) : x_1 > 0, x_2 > 0\}$

(d).  $f(x_1, x_2) = x_1x_2^{-1/2}$  on  $\mathbb{R}_{++}^2 = \{(x_1, x_2) : x_1 > 0, x_2 > 0\}$

(e).  $f(x_1, x_2) = x_1^\alpha x_2^{1-\alpha}$ , where  $0 \leq \alpha \leq 1$ , on  $\mathbb{R}_{++}^2 = \{(x_1, x_2) : x_1 > 0, x_2 > 0\}$

*Solution.* (a). This is a quadratic function  $f(\mathbf{x}) = \mathbf{x}^T \mathbf{A} \mathbf{x}$  with

$$\mathbf{A} = \begin{bmatrix} 2 & 0 & 1 \\ 0 & 2 & 1 \\ 1 & 1 & 1 \end{bmatrix}$$

The characteristic polynomial is

$$\det(\lambda \mathbf{I} - \mathbf{A}) = \det \begin{bmatrix} \lambda - 2 & 0 & -1 \\ 0 & \lambda - 2 & -1 \\ -1 & -1 & \lambda - 1 \end{bmatrix} = \lambda(\lambda - 2)(\lambda - 3)$$

The eigenvalues are 0, 2, 3, all nonnegative. So  $\mathbf{A}$  is positive semidefinite and  $f$  is convex.

(b). The Hessian matrix is

$$\mathbf{H} = \nabla^2 f(\mathbf{x}) = \begin{bmatrix} \frac{2}{x_1^3 x_2} & \frac{1}{x_1^2 x_2^2} \\ \frac{1}{x_1^2 x_2^2} & \frac{2}{x_1 x_2^3} \end{bmatrix}$$

Check the leading principal minors.

$$D_1(\mathbf{H}) = \frac{2}{x_1^3 x_2} > 0, \quad D_2(\mathbf{H}) = \det \mathbf{H} = \frac{3}{x_1^4 x_2^4} > 0$$

So  $\mathbf{H}$  is positive definite and  $f$  is convex.

(c). The Hessian matrix is

$$\mathbf{H} = \nabla^2 f(\mathbf{x}) = \begin{pmatrix} 0 & 2x_2 \\ 2x_2 & 2x_1 \end{pmatrix}$$

Since

$$\det \mathbf{H} = \det(-\mathbf{H}) = -4x_2^2 < 0,$$

$\mathbf{H}$  is neither positive semidefinite nor negative semidefinite, and  $f$  is neither convex nor concave.

(d). The Hessian matrix is

$$\mathbf{H} = \nabla^2 f(\mathbf{x}) = \begin{bmatrix} 0 & -\frac{1}{2}x_2^{-3/2} \\ -\frac{1}{2}x_2^{-3/2} & \frac{3}{4}x_1x_2^{-5/2} \end{bmatrix}$$

Since

$$\det \mathbf{H} = \det(-\mathbf{H}) = -\frac{1}{4x_2^3} < 0,$$

$\mathbf{H}$  is neither positive semidefinite nor negative semidefinite, and  $f$  is neither convex nor concave.

(e). The Hessian matrix is

$$\mathbf{H} = \nabla^2 f(\mathbf{x}) = \begin{bmatrix} \alpha(\alpha-1)x_1^{\alpha-2}x_2^{1-\alpha} & \alpha(1-\alpha)x_1^{\alpha-1}x_2^{-\alpha} \\ \alpha(1-\alpha)x_1^{\alpha-1}x_2^{-\alpha} & \alpha(\alpha-1)x_1^\alpha x_2^{-\alpha-1} \end{bmatrix} = -\alpha(1-\alpha)x_1^{\alpha-2}x_2^{-\alpha-1} \begin{bmatrix} x_2^2 & -x_1x_2 \\ -x_1x_2 & x_1^2 \end{bmatrix}$$

Check the principal minors of  $\mathbf{A} = \begin{bmatrix} x_2^2 & -x_1x_2 \\ -x_1x_2 & x_1^2 \end{bmatrix}$ .

$$\det \mathbf{A}_{11} = x_2^2 > 0, \quad \det \mathbf{A}_{22} = x_1^2 > 0, \quad \det \mathbf{A} = 0$$

So  $\mathbf{A}$  is positive semidefinite. Since  $\mathbf{H}$  is a negative multiple of  $\mathbf{A}$ ,  $\mathbf{H}$  is negative semidefinite and  $f$  is concave. Alternatively, we can see the positive semidefiniteness from the following representation,

$$\mathbf{A} = \begin{bmatrix} x_2 \\ -x_1 \end{bmatrix} \begin{bmatrix} x_2 \\ -x_1 \end{bmatrix}^T \succeq \mathbf{O}$$

□

4. Suppose  $f_i : \mathbb{R} \rightarrow \mathbb{R}$ ,  $i = 1, 2$ , are strictly convex functions. Show that  $f : \mathbb{R}^2 \rightarrow \mathbb{R}$  defined by  $f(x_1, x_2) = f_1(x_1) + f_2(x_2)$  is strictly convex over  $\mathbb{R}^2$ , and in particular  $f(x_1, x_2) = x_1^2 + x_2^4$  is strictly convex.

*Proof.* Let  $\mathbf{x} \neq \mathbf{y} \in \mathbb{R}^2$  and  $\theta \in (0, 1)$ . By the strict convexity of  $f_1$ ,

$$f_1(\theta x_1 + \bar{\theta} y_1) \leq \theta f_1(x_1) + \bar{\theta} f_1(y_1)$$

with strict inequality if  $x_1 \neq y_1$ . Similarly,

$$f_2(\theta x_2 + \bar{\theta} y_2) \leq \theta f_2(x_2) + \bar{\theta} f_2(y_2)$$

with strict inequality if  $x_2 \neq y_2$ . Since  $\mathbf{x} \neq \mathbf{y}$ , at least one of the above inequalities is strict. Adding them together yields

$$\begin{aligned} f(\theta \mathbf{x} + \bar{\theta} \mathbf{y}) &= f_1(\theta x_1 + \bar{\theta} y_1) + f_2(\theta x_2 + \bar{\theta} y_2) \\ &< \theta f_1(x_1) + \bar{\theta} f_1(y_1) + \theta f_2(x_2) + \bar{\theta} f_2(y_2) \\ &= \theta f(\mathbf{x}) + \bar{\theta} f(\mathbf{y}) \end{aligned}$$

Therefore,  $f$  is strictly convex. Since  $f_1(x) = x^2$  and  $f_2(x) = x^4$  are strictly convex,  $f(x_1, x_2) = x_1^2 + x_2^4$  is strictly convex. □

**Remark.** The result can be generalized as follows. For  $I \subset [n] \triangleq \{1, 2, \dots, n\}$ , let  $\pi_I : \mathbb{R}^n \rightarrow \mathbb{R}^{|I|}$  denote the projection on the coordinates in  $I$ , where  $|I|$  is the number of elements in  $I$ . For example, if  $n = 4$  and  $I = \{1, 3\}$ , then  $\pi_I(\mathbf{x}) = (x_1, x_3)$  for  $\mathbf{x} = (x_1, x_2, x_3, x_4) \in \mathbb{R}^4$ . Suppose  $f_i : \mathbb{R}^{n_i} \rightarrow \mathbb{R}$ ,  $i = 1, 2, \dots, k$  are strictly convex functions with  $n_i \leq n$ . If  $I_1, \dots, I_k \subset [n]$  such that  $|I_i| = n_i$  and  $\bigcup_{i=1}^k I_i = [n]$ , then  $f(\mathbf{x}) = \sum_{i=1}^k f_i(\pi_{I_i}(\mathbf{x}))$  is strictly convex.

5. Let  $f : C \subset \mathbb{R}^n \rightarrow \mathbb{R}$  be a differentiable function defined on a nonempty open convex set  $C$ . Show that  $f$  is convex if and only if

$$\langle \nabla f(\mathbf{x}) - \nabla f(\mathbf{y}), \mathbf{x} - \mathbf{y} \rangle \geq 0, \quad \forall \mathbf{x}, \mathbf{y} \in C. \quad (1)$$

Hint: For the sufficiency, consider the restrictions of  $f$  to straight lines, and note that a univariate function  $h$  is increasing iff  $[h(t) - h(s)](t - s) \geq 0$ . You can assume the fact that the intersection of  $C$  with a straight line is an open interval when it is not empty.

*Proof.* “ $\Rightarrow$ ”. Assume  $f$  is convex. By the first-order condition,

$$f(\mathbf{y}) \geq f(\mathbf{x}) + \langle \nabla f(\mathbf{x}), \mathbf{y} - \mathbf{x} \rangle$$

$$f(\mathbf{x}) \geq f(\mathbf{y}) + \langle \nabla f(\mathbf{y}), \mathbf{x} - \mathbf{y} \rangle$$

Adding them together yields

$$\langle \nabla f(\mathbf{x}) - \nabla f(\mathbf{y}), \mathbf{x} - \mathbf{y} \rangle \geq 0$$

“ $\Leftarrow$ ”. Assume (1) holds. Let  $\mathbf{x} \in \text{dom } f$  and  $\mathbf{d} \in \mathbb{R}^n$ . Let  $\mathbf{x}_t = \mathbf{x} + t\mathbf{d}$  and  $g(t) = f(\mathbf{x}_t)$ . Assume  $\text{dom } g \neq \emptyset$ , in which case it is an open interval. By the chain rule,

$$g'(t) = \langle \nabla f(\mathbf{x}_t), \mathbf{d} \rangle, \quad g'(s) = \langle \nabla f(\mathbf{x}_s), \mathbf{d} \rangle.$$

Note  $\mathbf{x}_t - \mathbf{x}_s = (t - s)\mathbf{d}$ . By (1),

$$[g'(t) - g'(s)](t - s) = \langle \nabla f(\mathbf{x}_t) - \nabla f(\mathbf{x}_s), (t - s)\mathbf{d} \rangle = \langle \nabla f(\mathbf{x}_t) - \nabla f(\mathbf{x}_s), \mathbf{x}_t - \mathbf{x}_s \rangle \geq 0,$$

so  $g'(t)$  is increasing on  $\text{dom } g$ , and  $g$  is convex. Since  $\mathbf{x}$  and  $\mathbf{d}$  are arbitrary,  $f$  is convex. □